

Community-acquired pneumonia in adults: acute and long-term complications

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Annually, community-acquired pneumonia (CAP) affects millions of individuals worldwide, resulting in more than 200 000 hospital admissions in England alone. CAP is a major cause of morbidity and mortality. Previous studies have explored the causes, risk factors, and the clinical course of CAP in detail. Sepsis and respiratory failure are important well-recognised consequences of progressive infection and have been researched extensively. In this Review, we focus on complications of CAP beyond the natural history of the condition. These sequelae of CAP often have long-term implications for patients and encompass a wide range of physical, functional, and psychosocial impairments, including reduced functional ability and persistent symptoms that adversely affect quality of life. Research on complications after CAP has increased since the COVID-19 pandemic. A growing body of literature informs clinical trials and guides clinical management. We explore the recovery trajectory and discuss updates in the management of post-CAP pleural infection, the established risk of cardiovascular events after CAP, and association of CAP with cognitive impairment and lung cancer. Finally, we conclude that improved long-term data, standardisation, and targeted research are needed to understand, prevent, and manage these outcomes effectively.

Introduction

Community-acquired pneumonia (CAP) is an infection of the lung parenchyma acquired outside hospital. CAP is a major worldwide health-care burden: annual incidence is 400–1100 per 100 000 adults,^{1,2} 22–42% of people with CAP are admitted to hospitals,^{3,4} and up to 10% of those admitted require management in the intensive care unit (ICU).⁵ The rate of readmission to the hospital within 30 days of discharge is 15–25%;^{6,7} most individuals are readmitted with respiratory issues, indicating a long-term clinical course with protracted morbidity.⁸ Mortality is also substantial: 5–15% of those admitted to hospitals with CAP die within 30 days, increasing to up to 50% in those admitted to the ICU.⁹ Importantly, mortality rates remain elevated years after the acute infectious period,¹⁰ suggesting that CAP predisposes individuals to long-term complications and adverse health outcomes. People who are old, frail, or comorbid are disproportionately affected, with increased incidence of complications, hospital admissions, and mortality. In the context of an increasingly ageing and comorbid population, complications of CAP demand improved consideration and exploration.

To systematically appraise the complications of CAP, the term complication requires a definition. However, no standard definition exists for a CAP complication in the literature; moreover, the term complication itself does not have a consensus definition. The Systematized Nomenclature of Medicine Clinical Terms defines a complication as an unexpected condition, outcome, or adverse event that occurs because of another condition, procedure, or treatment.¹¹ A 2025 statistical and linguistic analysis of more than 90 000 neurosurgical cases from Russia defined a complication as “an intercurrent pathology detected during observation of an underlying disease, physiological process, or the result of intervention”.¹² Notably, the authors did not define a complication based on cause-and-effect

relationships because of the complexity of interweaving pathological processes.

For the purpose of this Review, complications are defined as new or worsening conditions associated with CAP. Conditions arising from the initial pathological process or progressive infection, such as biochemical changes, respiratory failure, and sepsis, are not the focus of this Review. Instead, we explore evidence indicating pleural and cardiovascular complications, extended recovery, persistent symptoms, and diminished functional status. In addition, we review complications, such as cognitive impairment and lung cancer, that occur or persist years after the index pneumonia episode, although a causal relationship with CAP has not been established for these complications (figure 1). Studies have attempted to assess associations by adjusting for contributory factors such as age, frailty, and comorbidity. The ability of studies to distinguish reverse causality is poor.

Acute complications

Pleural complications

Pleural infection is a known complication of CAP and is considered to result primarily from the translocation of bacteria from the consolidated lung to the pleural space.¹³ Pleural effusion occurs in 20–57% of people with CAP,^{14,15} with 5–10% of patients with an effusion developing a pleural infection.^{15,16} The most common pathogens causing CAP-associated pleural infection are pneumococci, viridans streptococci, and *Staphylococcus aureus*,¹⁷ but increasing evidence indicates that most infections are polymicrobial.¹⁸ Pleural infection is a serious complication, with approximately 15% of individuals with pleural infection requiring thoracic surgery^{19,20} and a 1-year mortality rate of 20%.²¹ These factors create a substantial burden on health-care services, with an average hospital length of stay

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Key messages

- Pleural infection is an uncommon complication of community-acquired pneumonia (CAP) that confers high mortality. Intrapleural fibrinolytic therapy and immunomodulatory medications are being explored to improve outcomes.
- CAP increases the risk of cardiovascular events by activating atherosclerotic plaque inflammatory cells, cardiac remodelling, and inducing a prothrombotic state. Trials evaluating the use of antiplatelet therapy in reducing cardiovascular risk are under way.
- A third of the patients admitted to hospital with CAP experience functional decline in hospital; patients who are older, frailer, and have more comorbidities are more susceptible. Acute sarcopenia might also delay physical recovery in patients with CAP.
- More than half of the patients admitted to hospital with CAP subsequently present to primary care within 30 days of discharge. The most common reason for primary care consultation is persistent symptomatology.
- Moderate-to-severe cognitive impairment affects one in four people admitted to hospital with CAP. Hospitalisation for non-pneumonia infection confers a similar risk, suggesting an acute inflammatory aetiology.
- Rates of lung cancer diagnosis increase after CAP, even among individuals who have never smoked. Excluding patients diagnosed within a latency exclusion period to account for underlying malignancy does not eliminate this increased risk.
- Burden of CAP is higher in low-income and middle-income countries than in high-income countries, with increased rates of pulmonary tuberculosis. Individuals misdiagnosed with non-tuberculous pneumonia might be at increased risk of complications.

of 14 days.²² Concerningly, the incidence of pleural infection is rising, especially adults older than 60 years. One hypothesis for increased pleural infection rates is increased immunosuppression because of immunosuppressive agents and diabetes.²² Another explanation is increased detection because of greater use of CT. Overdetection is less likely given that outcomes, including mortality, surgical rates, and length of stay, have remained consistent; improved outcomes would be anticipated if more minor cases were detected.

Predicting which individuals with CAP will develop pleural infection remains challenging. Alcohol excess, intravenous drug use, thrombocytosis, hypoalbuminaemia, and hyponatraemia are known predictors of adult pleural infection; however, no validated risk score exists to identify individuals at risk. Poor dental hygiene has also been implicated as a risk factor for pleural infection because of microaspiration, with bacteria from the oral microbiota identified in up to 82% of pleural infection cases.²³

In 2023, the British Thoracic Society (BTS) and European Respiratory Society published similar guidelines on the management of pleural infection.^{24,25} Key updates included the use of dual intrapleural antifibrinolytic therapy to improve drainage of septated effusions¹⁹ and early surgical referral if no improvement occurs within 48 h. BTS guidance also highlights the rapidly assessed predictor of intraoperative damage score,²⁶ which includes age, urea, albumin, hospital-acquired infection, and non-purulence of fluid; clinicians can use this score to stratify mortality risk in confirmed cases of pleural infection.²⁰ Several randomised controlled trials (RCTs) are anticipated to further evaluate current management strategies. The multicentre intrapleural sepsis trial-4 study will compare clinical outcomes for intrapleural fibrinolytic therapy versus video-assisted thoracic surgery after initial treatment with antibiotics and chest tube insertion.²⁷ The initial management of pleural infection: aspiration versus chest tube trial aims

to assess the effect of therapeutic aspiration versus chest tube insertion on hospital length of stay.²⁸

New directions being explored to manage pleural infection include the use of immunomodulatory drugs.²⁹ The Corticoids for Pleural Effusion and Empyema (CORTEEC) RCT in children showed that intravenous dexamethasone reduced the time to clinical recovery (from 177 h to 109 h; $p=0.037$) but was underpowered to assess long-term or life-threatening outcomes.³⁰ In adults, although a feasibility study of intravenous dexamethasone use has been performed (the steroid therapy and outcome of parapneumonic pleural effusions [STOPPE] trial),³¹ a full-scale trial is awaited. Pleural interleukin-6, which is associated with hospital length of stay in pleural infection, is also a potential therapeutic target.³²

Cardiovascular complications

Pneumonia is increasingly being associated with a high short-to-medium term risk of major adverse cardiovascular events (MACE), such as myocardial infarction or stroke.³³ The physiological mechanisms underlying this risk are multifactorial. First, the inflammatory response to pneumonia activates inflammatory cells in atherosclerotic plaques, reducing plaque stability.³⁴ Second, acute infections can lead to platelet activation and a prothrombotic state persisting for several months, further predisposing susceptible individuals (ie, those with cardiovascular risk factors) to MACE.³⁵ Finally, hypoxic stress and increased metabolic demands secondary to pneumonia predispose individuals to arrhythmia, which in turn leads to cardiac ischaemia, especially in individuals with underlying cardiac disease.³⁶ The absolute risk of MACE for an individual depends on the severity of pneumonia and baseline cardiovascular risk. Estimated MACE incidence is 1–2% in individuals treated in primary care settings, 5–10% in individuals treated in hospitals, and 15% in patients who required ICU admission.³⁷ Incidence peaks in the first weeks following infection but does not return to baseline

for several months or even years.³⁸ Furthermore, interventions such as vaccination against respiratory pathogens, including influenza and pneumococcus, reduce cardiovascular events, presumably through reduced incidence and severity of corresponding infections.^{39,40}

Streptococcus pneumoniae is the most commonly associated pathogen with MACE, affecting up to one in five patients admitted to the hospital with pneumococcal pneumonia.⁴¹ Patients with concurrent MACE have significantly higher mortality than those with pneumococcal pneumonia alone (odds ratio [OR] 3.9; $p < 0.008$), emphasising the importance of understanding underlying mechanisms. Evidence indicates cardiac remodelling and scarring after pneumococcal infection.⁴² *S pneumoniae* can cause cardiac microlesions and atherosclerotic plaque rupture through direct translocation into the myocardium.⁴³ Analyses suggest that pneumolysin, a pore-forming toxin, might mediate cardiotoxicity. Serum troponin concentrations correlate with prognosis in individuals who are critically ill with pneumonia in the absence of MACE.⁴⁴ Pneumococcus is not the only pathogen associated with increased risk of MACE. Mice infected with influenza A have viral RNA in their arterial walls, with a higher concentration of macrophages in atherosclerotic plaques, suggesting that infection might contribute to or exacerbate atherosclerosis.^{45,46}

In addition to infection-related risk, pharmacological treatment might affect MACE risk. Macrolides are widely administered for acute treatment in CAP and infection prophylaxis in chronic airways disease. Based on the hypothesis that clarithromycin might reduce the clinical manifestations of coronary atherosclerosis, the CLARICOR trial evaluated the effects of clarithromycin versus placebo over a 2-week period in individuals after an incidence of myocardial infarction or angina pectoris.⁴⁷ Cardiovascular mortality was significantly higher in the clarithromycin arm at both 2.6 years (hazard ratio [HR] 1.27 [95% CI 1.03–1.54]; $p = 0.024$) and 6 years (1.21 [1.06–1.38]; $p = 0.004$), raising concerns about potential cardiotoxicity. A 2017 systematic review and meta-analysis of the association between macrolides and cardiovascular risk across 41 studies identified a short-term increased risk of myocardial infarction in cohort studies (relative risk [RR] 3.53 [95% CI 2.25–5.54]) but not in RCTs (0.65 [0.24–1.75]).⁴⁸ Notably, no increased short-term risk of arrhythmia or stroke or long-term risk (> 30 days) was identified. A 2026 post-hoc analysis of macrolide use in patients hospitalised with CAP reported a trend towards increased risk of cardiovascular events (OR 1.41 [95% CI 0.92–2.16]) but decreased risk of mortality (0.94 [0.50–1.76]) in those receiving macrolide therapy.⁴⁹ In 2023, an international task force published treatment guidelines for severe CAP,⁵⁰ although the potential risk of cardiotoxicity associated with macrolides was acknowledged in the guidelines, macrolides were recommended as part of combination therapy because of improved mortality.

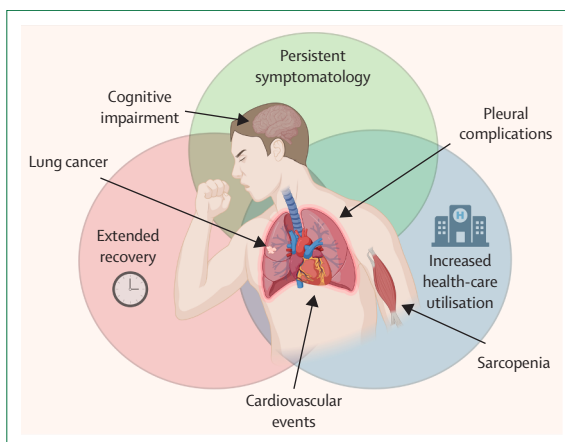


Figure 1: Complications of community-acquired pneumonia

Some complications of CAP are localised (arrows indicate approximate anatomical location), whereas others are diffuse and affect both the patient and the society (represented by coloured circles).

Evidence suggests that CAP can precipitate new-onset heart failure or arrhythmia in the absence of MACE. In an observational study of 5613 individuals aged 65 years or older, CAP was associated with a markedly increased risk of developing heart failure, both during and after treatment in the hospital.⁵¹ Those who survived at least 30 days after pneumonia-related hospital admission had a significantly increased risk of new-onset heart failure at days 30–90 (HR 6.9 [95% CI 4.46–10.63]; $p < 0.01$). This risk decreased over time but was still doubled at 5 years compared with that at baseline. The authors hypothesised that even in the absence of MACE, the cardiovascular stress and arrhythmogenic burden in clinically vulnerable individuals are sufficient to cause lasting damage. In a cohort of 50 119 predominately male US veterans who were aged 65 years or older and admitted to hospital with pneumonia, the risk of serious cardiac arrhythmias was 9.5% by day 90. Most arrhythmias occurred during hospitalisation, and the incidence was higher in those requiring admission to the ICU.⁵²

Several observational studies have compared cardiovascular outcomes among patients receiving antiplatelets; however, these studies were confounded by the indication for antiplatelet therapy. In 2021, a large population-based analysis of primary care records using propensity score matching observed that aspirin use was strongly associated with a reduced risk of myocardial infarction and stroke following pneumonia (adjusted HR 0.64 [95% CI 0.52–0.79]).³⁷ Using a similar methodology as that in the 2021 study, a 2024 study found reduced 28-day mortality in patients with CAP admitted to the ICU (HR 0.75 [0.63–0.88]; $p < 0.001$).⁵³ To date, only one small RCT has evaluated aspirin as an adjunct for CAP; aspirin therapy was associated with a 9% absolute risk reduction in MACE (RR 0.103 [95% CI 0.005–0.746]; $p = 0.015$).⁵⁴ Two RCTs—namely, ASPECT and ASPAPY—are under way with greater sample sizes than those in previous trials and are

expected to provide more definitive answers.^{55,56} Several other classes of drugs, including statins, angiotensin-converting enzyme inhibitors, and angiotensin receptor antagonists, have been studied for CAP but have not shown efficacy in reducing MACE or mortality.^{57,58}

Recovery from CAP

Symptomatology and functional ability

The multifactorial and extended effect of CAP on individuals often persists beyond clinical or radiographic resolution of the acute illness.⁵⁹ In a 2019 systematic review of patient-reported outcomes in adults who were admitted to hospital with CAP, a substantial proportion of participants reported persistent symptoms after 4–6 weeks of discharge; most frequently reported symptoms included fatigue (reported by 45.0–72.6% of participants), cough (35.3–69.7%), and dyspnoea (34.2–67.1%).^{60,61} Studies exploring interventions to improve symptomatology after CAP are scarce.

Health-related quality of life (HrQOL) is a useful marker of the consequence of a disease, as experienced by the patient.⁶² Long-lasting impairment in HrQOL is often reported after CAP. In a prospective matched cohort, HrQOL was significantly lower in the CAP cohort at 12 months across all domains of life quality, such as mobility, self-care, routine activities, pain or discomfort, and anxiety or depression.⁶³ The estimated 1-year excess quality-adjusted life year loss attributed to CAP was 0.13. Importantly, the burden of morbidity following CAP was not confined to older individuals; loss of ability to perform activities of daily living, reductions in HrQOL, and employment loss were also reported in studies with adults of working age who had CAP.⁶⁴

In addition to persistent symptomatology, individuals who had CAP often experience reduced functional ability. Over a third of individuals experience functional decline during a stay in hospital; the proportion of individuals affected increases with increasing duration of hospital stay.⁶⁵ In the context of long-term complications, functional decline might persist for over 6 weeks after discharge. Functional decline has been independently associated with functional status at admission, older age, comorbidities, ICU admission during hospital stay, and previous admissions to hospitals.⁶⁶ In a prospective cohort study of adults admitted to the hospital with CAP, functional independence was significantly lower at 45 days after discharge than before CAP onset, and 12.1% of individuals who previously lived in the community required institutionalisation with residential or long-term care after CAP.⁶⁵ Data assessing functional status in individuals who were admitted to the hospital for respiratory syncytial virus or influenza infection have shown a deterioration in functional independence similar to that in individuals with CAP.^{67,68} This observation suggests that pulmonary infection necessitating admission to hospital might instigate a trajectory of functional decline,

particularly in susceptible individuals—ie, those who are old, frail, and have comorbidities.⁶⁹

Sarcopenia

The association between CAP and functional decline might be partly explained by another complication of CAP—ie, sarcopenia. Sarcopenia is defined as a progressive and generalised skeletal muscle disorder, associated with adverse outcomes.⁷⁰ Sarcopenia increases susceptibility to pneumonia because of weakened oropharyngeal and respiratory musculature^{71,72} and is associated with worse outcomes following pneumonia.⁷³ Systemic inflammation and oxidative stress during pneumonia might cause muscle proteolysis, leading to sarcopenia.^{74,75} Muscle atrophy is widely recognised in the intensive care environment; patients who are critically ill lose approximately 2% of their muscle mass per day.⁷⁶ Recognised causes of sarcopenia include systemic inflammation, immobilisation, and malnutrition,⁷⁷ all of which are likely to affect patients admitted to the hospital with CAP or other acute infections.

Longitudinal data examining the development of sarcopenia after CAP are scarce; however, small-scale studies suggest that sarcopenia might arise acutely. In an RCT investigating the effect of an 8-day inpatient physical training regimen versus traditional respiratory physiotherapy, the control group showed muscle deterioration, whereas the intervention group showed improvement across all muscle groups from baseline after intervention.⁷⁸ Additionally, in a study of 62 patients who were aged 65 years or older with severe CAP, acute loss of skeletal muscle mass independently predicted 90-day mortality after controlling for pre-existing frailty and pneumonia severity (adjusted HR 1.40 [95% CI 1.137–1.725]; $p=0.002$).⁷⁹ Although larger-scale studies with longer follow-up periods are needed, the RCT investigating physical training and the study of 62 older patients, both suggest that CAP might increase the risk of acute muscle atrophy. A complex interplay exists between sarcopenia, frailty, and malnutrition, and interventions targeting these complications might improve longer term poor outcomes (eg, hospital readmissions, morbidity, and mortality) experienced by these individuals.

Health-care utilisation

Standardising morbidity outcomes in CAP research remains challenging because of the complexity and heterogeneity of recovery after illness. A potential surrogate marker is health-care utilisation. Importantly, there is potential for survivorship bias; most studies review admission to hospital data, implying that individuals must be considered sufficiently fit to benefit from hospital admission to be included in these studies. This criterion disproportionately excludes those with terminal illness, severe frailty, and do-not-hospitalise directives. In a cohort study of almost 40 000 individuals with CAP in England,

	Population	Relevant outcome measure	Key findings
UK retrospective cohort study ⁸	8136 adults admitted to the hospital with CAP during 2018–19	Readmission rates and reasons	30-day readmission rate of 15.7% 61.3% of readmissions for respiratory reasons; 9% of readmissions for circulatory reasons In-patient mortality was 15.9% in those readmitted due to pneumonia, compared with 6.5% in those readmitted with other diagnoses 3.4% required a new residential care placement on discharge Follow-up planned in 71.6% of the participants after discharge Readmission highest in older people, individuals with comorbidities, and those with severe CAP on index admission
Danish retrospective cohort study ⁸¹	246 245 older adults (aged 65–99 years) admitted to hospital with CAP during 2000–16	Readmission rates; mortality	30-day readmission rate of 23.0% 30-day mortality of 8.1% Readmission highest in older people and individuals with higher number of past admissions
US retrospective cohort study ⁸²	37 006 adults (with a mean age of 71 years) admitted to the hospital with pneumonia during 2012–18	Readmission rates	30-day readmission rate of 12.5% 360-day readmission rate of 42.3% Readmission highest in older people and individuals with high-risk conditions, such as chronic renal failure, HIV, active malignancy, or undergoing immunotherapy
UK retrospective cohort study ⁸³	56 396 adults admitted to the hospital with pneumonia during 2002–17	Consultation with general physician after discharge from the hospital	55.9% adults consulted with general physician within 30 days of discharge Consultation highest within 7 days of discharge 47.7% of the adults consulted with general physician more than one time Among those who reconsulted primary care within 30 days of discharge, 40.7% did so for a respiratory issue 31.1% received additional antibiotic prescriptions Reconsultation rates highest in adults with a greater number of primary care consultations before index pneumonia diagnosis, a history of smoking, existing cardiac and respiratory conditions, or existing diabetes
Systematic review ⁸⁴	17 studies with most of the retrospective, 3 million participants	Risk factors associated with readmission	Higher rates of readmission associated with male sex, malignancy, heart failure, chronic kidney disease, chronic respiratory disease, and diabetes
UK prospective cohort study ⁸⁵	108 patients (with a median age of 50 years) admitted to the hospital with pneumonia during 2015–16	Health-care reconsultation after discharge from the hospital by adults of working age	65.7% had a health-care consult within 28 days of discharge 90.1% consulted health care through primary care; 18.3% consulted via emergency services 4.6% of patients readmitted within 4 weeks
Taiwanese single-centre RCT ⁸⁶	82 older adults (older than 65 years) who were malnourished and admitted to the hospital with pneumonia during 2017–18	Effect of nutritional intervention on readmission rates	77% reduction in readmission for pneumonia within 6 months in the intervention arm compared with the control group
Spanish multicentre RCT ⁸⁷	102 adults admitted to the hospital with CAP during 2011–14	Effect of educational intervention at discharge on health-care consultation after treatment in the hospital	23.5% of the adults who received personalised education had either an additional health-care visit or rehospitalisation compared with 42.9% of the adults in the control arm
CAP=community-acquired pneumonia. HIV=human immunodeficiency virus. RCT=randomised controlled trial.			
Table: Summary of studies related to health-care utilisation after admission to hospitals for CAP			

individuals in residential care or with frailty factors, including bedsores, poor nutritional status, and incontinence, were less likely to be admitted for pneumonia.⁸⁰ The bias is likely more pronounced when considering readmission data, in which individuals must survive their index admission and remain acceptably fit for readmission to be included in the statistics. Keeping these limitations in view, health-care utilisation data associated with individuals admitted to hospitals for CAP remain an important marker of morbidity (table).

Long-term complications

Although acute complications are easier to be linked with CAP, as these complications occur during or immediately after the index pneumonia episode, long-term complications are more complex to be attributed to CAP. Individuals who develop CAP are older, frailer, and have more comorbidities than those who do not develop CAP;⁸⁸ thus,

distinguishing whether long-term sequelae are attributable to CAP or underlying health trajectories is difficult. Additionally, individuals admitted to the hospital for CAP might have pre-existing undiagnosed conditions, which remain detected during or following admission (figure 2).

Cognitive impairment

Up to one in four people have new, moderate-to-severe cognitive impairment persisting for a year after treatment in hospital with CAP, and an additional one in three people have mild cognitive impairment.⁸⁹ The risk of cognitive impairment and dementia is higher in people hospitalised with pneumonia than in matched controls (adjusted HR 2.24–2.5 in prospective cohort studies;^{90,91} 1.10–1.88 in retrospective cohort studies^{92,93}). A higher incidence of cognitive impairment is reported with more recent pneumonia-related admission to hospitals.^{93,94} Whether cognitive decline is persistent is debatable; a longitudinal

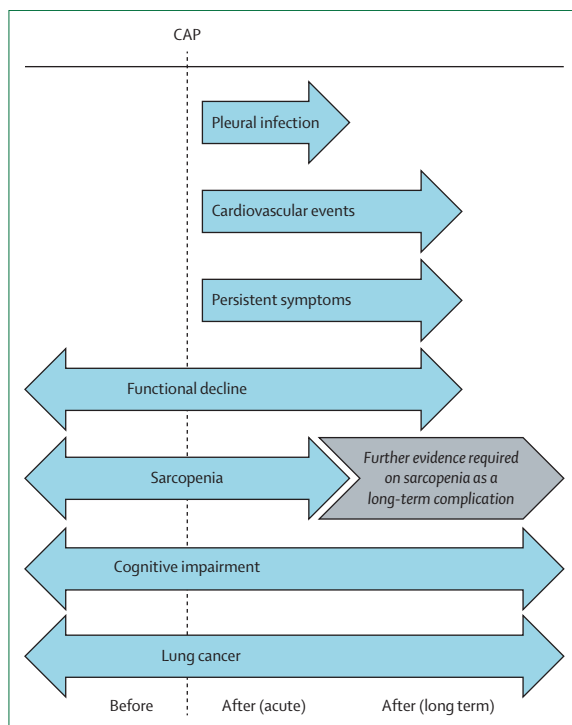


Figure 2: Complications of CAP over time

Pleural infection is an acute complication, whereas cardiovascular events occur in the short term and medium term following CAP. Symptoms of CAP might persist during the short and medium term. Functional decline, sarcopenia, cognitive impairment, and lung cancer have a bidirectional relationship with CAP; each condition might predispose an individual to CAP, whereas CAP might also worsen or exacerbate the underlying condition. Evidence indicating long-term incidence of sarcopenia is scarce. Functional decline persists into the medium term, whereas cognitive impairment and increased lung cancer diagnosis persist into the long term. CAP=community-acquired pneumonia.

cohort study noted only a transient effect on cognitive decline over the first 2.5 years from diagnosis,⁹⁴ whereas other studies have reported a persistent effect.^{89,93,95}

Although individuals aged 65 years or older have the highest incidence of cognitive impairment following pneumonia-related admission to hospitals,^{89,93} 19% of survivors of pneumonia younger than 65 years had moderate-to-severe cognitive impairment 12 months after a pneumonia diagnosis.⁸⁹ This observation suggests that cognitive impairment is not confined to older adults. When analysed by age range, the adjusted HR for cognitive impairment or dementia was greatest in 45–60-year-olds, likely reflecting the low baseline rate of cognitive decline in this group.⁹³

An increased incidence of cognitive impairment in older survivors of pneumonia might be attributed to pre-existing cognitive trajectories. In a population cohort study,⁹⁰ pre-existing subclinical changes in cognition increased the risk of pneumonia. A mini-mental state examination score lower by ten points (score 0–100) was associated with a 7.7% increased hazard of developing pneumonia.⁹⁰ People with dementia also have an increased incidence of pneumonia;⁹⁶ subclinical cognitive impairment might confer a similar, albeit attenuated, risk. Physical health might also

play a role; in a longitudinal study of almost 10 000 individuals, dementia risk was increased in survivors of pneumonia with a history of pre-frailty or frailty but not in those without frailty,⁹⁷ whereas numerous other studies have shown an increased risk of cognitive impairment in survivors of pneumonia with comorbidities.^{89,92,93} In all studies, dementia risk persisted after adjustment for comorbidities.

Admission to hospital for non-pneumonic infection confers a risk of dementia similar to that conferred by pneumonia-related admission to hospital.^{90–92,98} In contrast, hospitalisation without infection did not increase the risk of dementia,^{90,91} suggesting that the infectious process itself might be responsible for the increased risk. Delirium occurs in one in five patients hospitalised with pneumonia; rates do not vary significantly between microbiological entity or pneumonia severity but are increased in patients requiring ICU admission.⁹⁹ Acute systemic inflammation induces neuroinflammation through microglial activation or priming;¹⁰⁰ in mouse models, such neuroinflammation leads to behavioural deficits, consistent with delirium.¹⁰¹ In some individuals, microglial overactivation might lead to uncontrolled neuroinflammation and neurodegeneration, resulting in extended cognitive impairment and dementia.¹⁰² The risk of dementia is eight-times higher in individuals who have had delirium,¹⁰³ with a four-times increased risk of mortality following pneumonia.⁹⁹ Further understanding of the association between pneumonia and cognitive impairment and exploration of preventive approaches, such as targeting modifiable risk factors for dementia in those at highest risk, is crucial to improving debilitating long-term cognitive impairment.

Lung cancer

Numerous case-control and population-based cohort studies have shown an association between a previous pneumonia episode and lung cancer in non-smoking women with lung cancer self-reporting a history of pneumonia (OR 1.37 [95% CI 1.0–1.8]) and in women who smoked and were hospitalised with pneumonia (16.35 [11.57–23.11]; $p<0.001$).^{104–108} In a 2011 systematic review and meta-analysis on previous lung disease and lung cancer, the RR from 22 studies was 1.43 (1.22–1.68), and the risk remained elevated when the analysis was restricted to people who had never smoked.¹⁰⁹ A subsequent pooled analysis of studies in the International Cancer Consortium showed an RR of 1.57 (1.22–2.01, 12 studies).¹¹⁰ Lung cancer incidence is higher with a more recent pneumonia diagnosis, with rates of 1.7–8.14% within the first year.^{106,111–114} Of note, in case-control studies of people with lung cancer, in which exposure risk is dependent on personal recall of preceding pneumonia, recall bias is likely to affect results.

Rates of pneumonia are higher in those with lung cancer, likely because of obstructive lesions and immune dysregulation. Persistent lung cancer, presenting as pneumonia, has been theorised to account for some of the observed associations between lung cancer and pneumonia. To

adjust for potential underlying malignancy, studies have introduced latency exclusion periods into their analyses, ranging from 30 days to 3 years. However, elevated risk remains for those with lung cancer.^{104,105,107} Median time from pneumonia to lung cancer diagnosis is 109–297 days.^{111,113} Studies with prolonged follow-up reported ongoing increased risk for up to 6 years following pneumonia diagnosis.^{106,111} Two studies assessed the association between pneumonia and lung cancer incidence by histological subtype, and both reported an increased OR for non-adenocarcinoma (adjusted OR 1.71–2.1) compared with that for adenocarcinoma (adjusted OR 1.1–1.31).^{104,108} In a 2025 study of quiescent metastatic breast cancer in a mouse model, infection with influenza A triggered dormant cancer cell reactivation and proliferation in the lungs through proinflammatory and angiogenic signalling cascades.¹¹⁵ Despite the use of only an early animal model, the results introduce a potential mechanism by which respiratory infection might trigger cancer cell proliferation.

Factors associated with an increased risk of lung cancer diagnosis in survivors of pneumonia include increasing age, male sex, current or former smoking, history of malignancy, and presence of comorbidities, including previous lung disease and congestive heart failure.^{108,111,113} Increased health-care interactions in those with comorbidities, including more imaging and health-care interactions, might lead to lead time ascertainment bias and overdiagnosis. Smoking causes both pneumonia and lung cancer, making it challenging to establish an independent causal relationship. In non-smoking women who have previously had pneumonia, after adjusting for demographics and previous lung disease, an attenuated but persistently increased risk of lung cancer diagnosis was observed (adjusted OR 1.22 [95% CI 0.9–1.6]). Possible mechanistic explanations include post-pneumonic chronic inflammation or impaired healing.^{116,117} Individuals prone to pneumonia are also prone to lung cancer through impaired airway clearance¹¹⁸ or dysregulated immune function.

The value of repeat or follow-up imaging to assess for underlying malignancy in survivors of pneumonia is unclear. The BTS recommends a repeat chest x-ray in 6–8 weeks for those at higher risk of malignancy; however, the evidence underlying this recommendation is weak (grade D).¹ The National Institute of Health and Care Excellence (NICE) has recently advised against routine follow-up chest x-rays for those discharged from inpatient care after an episode of CAP, unless the patients are at high risk of lung cancer or have persistent symptoms.¹¹⁹ NICE recognised the uncertainty by recommending research on the clinical and cost-effectiveness of follow-up imaging in pneumonia as a priority. The American Thoracic Society advises against a routine follow-up chest x-ray in those whose symptoms have resolved within 5–7 days.^{1,120}

Studies from Canada and New Zealand report that 40–53% of survivors of pneumonia undergo a follow-up chest x-ray within 90 days; diagnostic yield for cancer in

those who underwent follow-up imaging was 2.5–3.7%, and the rates were higher in individuals who were aged 50 years or older.^{111,114} In NHS England's lung cancer screening programme, individuals are considered at high risk if their calculated lung cancer risk prediction scores are 1.51% or higher over 6 years of using the prostate, lung, colorectal and ovarian cancer (PLCOm₂₀₁₂) model or are 2.5% or higher over 5 years of using the Liverpool lung project (LLPv2) model.^{121–123} In this regard, a yield of 2–2.5% on follow-up imaging could be considered sufficient to justify follow-up imaging. As low-dose CT scans become increasingly accessible, the value of chest x-ray imaging to detect lung cancer should be examined. In a study of over 2000 patients with lung cancer who had a chest x-ray in the preceding year, diagnostic sensitivity was only 82.3% (95% CI 80.6–84.1%),¹²⁴ resulting in both underdetection and false reassurance. Other studies report similar diagnostic accuracy.¹²⁵ Variations in access to low-dose CT scanning should be considered, particularly in the context of health-care settings with minimal resources.

CAP complications in other health-care settings

Most studies have been conducted in high-income countries, in patients who were hospitalised; however, the results of these studies are not representative of the global burden of disease. Most adults with CAP receive care in the community, and low-income and middle-income countries (LMICs) have more CAP diagnoses than high-income countries. Research in LMICs should be prioritised to create a comprehensive understanding of CAP complications.

CAP management in the community

The scarcity of literature on the complications of CAP in adults who were treated as outpatients is partly because of difficulties with diagnoses; in the community, many individuals are diagnosed clinically, without chest imaging. In addition, those admitted to hospitals with pneumonia are likely to be more symptomatic, have more severe infections, and undergo more frequent investigations than those not admitted; thus, individuals admitted to hospitals are not representative of those who receive care in the community. Analysis of coded primary care data has helped to understand the cardiovascular risk associated with CAP.³⁷ Increased data availability and integration of primary and secondary care databases will facilitate further exploration of the complications of CAP in individuals who receive care within the community.

LMICs

LMICs experience a higher burden of acute CAP than high-income countries, with fewer resources to prevent, diagnose, or manage complications. Incidence of lower respiratory infections and associated mortality is higher in LMICs than in high-income countries, with a disproportionate increase in children younger than 5 years.¹²⁶ According to the 2019 Global Burden of Disease Study, 84%

Search strategy and selection criteria

Using Ovid, we searched MEDLINE and Embase for peer-reviewed articles published from 1946 to Jan 10, 2026, excluding studies involving children and animals, using a combination of indexing and free-text terms. We conducted a general search for CAP complications, associations, and risks. We also conducted separate searches for each identified complication (appendix). We then screened the retrieved articles for relevance and also considered additional studies known to the authors to be relevant.

See Online for appendix

of the approximately 2·49 million global deaths from lower respiratory infections occurred in LMICs (in which South Asia and sub-Saharan Africa were the most affected areas).¹²⁷ *S pneumoniae* remains the most common pathogen worldwide, followed by *Haemophilus influenzae* and viruses such as respiratory syncytial virus and influenza. People living with HIV, particularly those who are severely immunocompromised, might have a different spectrum of infections associated with CAP, such as pneumocystis pneumonia, tuberculosis disease, or opportunistic infections.^{128,129} Mortality from CAP is higher for people living with HIV than for those without HIV, although this rate has improved in recent years because of the introduction of pneumococcal vaccination and access to antiretroviral therapy.¹²⁶

Acute pulmonary tuberculosis, caused by *Mycobacterium tuberculosis*, accounts for 10–18% of CAP cases in LMICs.^{130,131} Pulmonary tuberculosis is common for several reasons: HIV is widespread and predisposes individuals to coinfection; BCG vaccination has lesser protective effect in adults than in children;¹³² and densely populated living areas facilitate transmission of the aerosol-borne infection. Pulmonary tuberculosis is often misdiagnosed as non-tuberculous CAP, owing to clinical and radiological similarities, suggesting a need to suspect pulmonary tuberculosis in acute CAP-like presentations.¹³¹ Atypical tuberculosis disease presentations might add a further diagnostic challenge. Given the different management approaches and the prevalence of multidrug-resistant tuberculosis disease, misidentification predisposes individuals to complications.

As with conventional CAP, pulmonary tuberculosis can cause long-term adverse effects on various organ systems. Lung disease developed after tuberculosis disease encapsulates several chronic respiratory abnormalities including parenchymal damage, airway disease, and pulmonary vascular disease; such lung disease occurs in approximately 50% of survivors of tuberculosis disease, with the severe form of lung disease developed in 10–15% of the survivors. Pulmonary tuberculosis is associated with cardiovascular complications: incidence of acute coronary syndrome was 40% higher in patients with tuberculosis disease than in patients without the disease, in a study with 13 years of follow-up.¹³³ A meta-analysis found cardiovascular disease to be the leading cause of mortality after tuberculosis treatment, accounting for 20% of all deaths.¹³⁴ Pulmonary tuberculosis is also associated with increased subsequent

lung cancer development in patients compared with that in control participants (incidence rate ratio 1·75 [95% CI 1·33–2·32]; $p < 0·001$).^{133,135} Socioeconomic impairment after tuberculosis disease is common owing to financial insecurity, physical morbidity, and stigma.¹³⁶ Pulmonary tuberculosis is associated with muscle weakness responsive to pulmonary rehabilitation,¹³⁷ increased health-care utilisation,¹³⁸ reduced wellbeing,¹³⁹ and mental health disorders with impaired cognitive functioning.¹⁴⁰ In view of the substantial morbidity and mortality from respiratory infections in LMICs, further research to reduce antimicrobial resistance and improve vaccination should be prioritised.¹⁴¹ A phase 3 clinical trial evaluating the M72/AS01E-4 vaccine for pulmonary tuberculosis is under way (ClinicalTrials.gov, NCT06062238), with early trials indicating efficacy and safety.^{142,143}

Conclusions and future directions

Complications of CAP have become increasingly recognised after the COVID-19 pandemic. Acute, short-term complications of CAP are more conclusively established, whereas long-term sequelae are still under evaluation. Long-term studies are more prone to confounding than short-term ones, making the elucidation of true associations difficult; yet, understanding long-term outcomes is essential. The expansion of digital health records, integration of primary and secondary care databases, and improved data processing technologies using artificial intelligence have facilitated large-scale analyses of CAP complications and associations. Importantly, electronic health-care records are often unavailable in settings with minimal resources; this limitation tends to widen the disparity in outcomes between high-income countries and LMICs. Hence, facilitating and developing research in LMICs should be prioritised.

Future research should focus on elucidating the characteristics of subgroups at greater risk of complications from CAP, and whether the risks are comparable between those admitted to hospitals and those receiving care in the community. Questions should focus on why some individuals experience persistent symptoms, what is the underlying mechanism connecting systemic infection and cognitive impairment, and whether individuals who are immunosuppressed experience different complications.

The underlying pathophysiology of complications is understood only partially, and the burden of complications is likely underestimated. Increasing clinician awareness and standardising definitions and reporting would help to evaluate the true prevalence. Similarly, standardised assessment tools to assess recovery from CAP are absent;¹⁴⁴ development of sensitive and specific measures that focus on patient-reported outcomes will improve understanding. Standardisation of both outcome frameworks and follow-up methodology will enhance the design of future studies. By accurately identifying the individuals at risk for complications, interventions can be implemented in a resource-efficient manner.

High-quality, long-term data assessing the effect of novel interventions on the trajectory of CAP complications are inadequate. Potential interventions under investigation include statins¹¹⁹ and glucocorticoids, both of which are associated with reduced CAP severity. Further large-scale RCTs are needed to clarify the utility of such interventions in reducing acute and long-term CAP complications. Given the substantial burden of CAP and the numerous complications associated with poor long-term health trajectories, recognising and mitigating the risk of complications should be an urgent research priority.

Contributors

JC and DA conceptualised the Review; JC, KB, RS, and JS conducted literature searches; JC and DA created the figures; JC, KB, RS, JS, and DA wrote the manuscript; FH, RP, GW, JS, JK, WSL, AB, NM, and DA provided expert opinions and reviewed the drafted manuscript; JC, RS, and DA edited the manuscript following Review; all authors were responsible for reviewing and approving the final manuscript.

Declarations of interest

We declare no competing interests.

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